CLINICAL AI COMPANION | AMD ROCM & VLLM ACCELERATED | ZERO HALLUCINATIONS

A Clear, Evidence-Linked Discharge Companion for Every Patient.

Eliminating post-hospital complications with private AI extraction, line-by-line verbatim grounding, cross-document conflict detection, and multilingual navigation.

TEAM
The Crawlers

AMD HARDWARE & STACK
ROCm 6.x • vLLM • Qwen2.5-7B-Instruct

LIVE DEPLOYMENT
carelynx-two.vercel.app

The Post-Hospital Discharge Crisis

Discharged patients receive dense, fragmented medical paperwork, creating severe clinical hazards when navigating recovery alone at home.

19–23%

ADVERSE MEDICAL EVENTS

Nearly 1 in 5 patients suffers a preventable complication within 30 days of leaving hospital care.

50%+

MEDICATION DISCREPANCIES

Over half of errors stem from misunderstood dosages, duplicate prescriptions, or forgotten drug changes.

65%

LINGUISTIC BARRIERS

Patients receive English papers with Latin abbreviations (`q.d.` , `b.i.d.` , `p.r.n.`) with no native translation.

$26B+

ANNUAL READMISSION BURDEN

Unplanned 30-day hospital readmissions impose billions in HRRP penalties on healthcare systems.

Paperwork Fragmentation:

A single patient encounter generates disparate documents: Clinical Discharge Summaries, Outpatient Prescriptions, Pathology Sheets, and Referral Letters — often with conflicting advice.

The Fundamental Dilemma:

"The critical medical information already exists in the paperwork, but patients cannot safely or clearly navigate it after leaving the hospital doors."

CARELYNX: Grounded Care Navigation

A safety-first clinical intelligence platform that ingests unstructured medical records and outputs verified, interactive patient care plans.

📄 100% Verbatim Evidence Grounding

Zero hallucinated clinical data. Every extracted medication, dose, allergy, and follow-up appointment links directly to line-by-line quotes and bounding boxes in the original hospital records. Patients click any fact to view original document proof.

⚠️ Deterministic Conflict Detection

Discharge documents frequently disagree (e.g. Discharge Note says *Lisinopril 10mg*, while Prescription Order says *20mg*). CARELYNX automatically cross-checks encounters, identifies discrepancies, and quarantines them from patient view.

👨‍⚕️ Human-in-the-Loop Clinical Review Queue

Ambiguous instructions, smudged OCR scans, or clinical conflicts are immediately escalated to a dedicated review workspace (`/review`) for discharge nurses and physicians to review and sign off.

🌐 Safe Multilingual Translation

Supports 6 languages (Hindi, Gujarati, Marathi, Tamil, Telugu, English). A strict **Dosage Preservation Covenant** keeps brand drug names and numerical doses untranslated while translating actionable instructions.

End-to-End System Architecture

Engineered for air-gapped on-premise hospital clusters or scalable cloud deployments with zero data leaks.

1

Ingest & OCR

Multi-page PDF / image upload. Magic-byte sniffing, size gates, and Tesseract 5.5 OCR. Computes page text-quality score (threshold ≥ 0.85).

2

AMD ROCm vLLM

High-throughput inference with `Qwen2.5-7B-Instruct` on ROCm 6.x. Continuous PagedAttention batching with structured JSON output schema.

3

Safety & Conflict

Deterministic conflict matrix. Verbatim substring validation. Auto-routes conflicting or low-confidence facts (<0.80) to review queue.

4

Care Plan & FHIR

Next.js 16 reactive dashboard. Calendar view, medication timeline, red-flag warning banners, and standard FHIR R4 Bundle export.

Core Technologies: FastAPI • SQLAlchemy • PostgreSQL / SQLite • Alembic • Next.js 16 (Turbopack) • TailwindCSS • Docker Compose • Tesseract 5.5

HIPAA-COMPLIANT STORAGE VAULT

Accelerated by AMD ROCm & vLLM

Leveraging AMD Instinct (MI300X/MI250) and Radeon hardware for high-throughput, private, local clinical intelligence.

⚡ vLLM PagedAttention on ROCm

Medical discharge records span 5 to 20 pages. Running vLLM on AMD ROCm delivers continuous batching with `gpu_memory_utilization=0.9` and `max_model_len=8192`, drastically minimizing TTFT for real-time extraction.

🔒 100% Private On-Premise AI

Zero Protected Health Information (PHI) ever leaves hospital infrastructure. Local execution on AMD Instinct hardware eliminates cloud API cost volatility and guarantees total data sovereignty.

📊 Audited Hardware Provenance

Every clinical extraction is stamped with verified hardware telemetry: `INFERENCE_HARDWARE_LABEL="amd-rocm-vllm"` via `rocm-smi` / `amd-smi` ASIC validation. Fully benchmarked via `benchmark_rocm.py`.

```
# Production AMD vLLM Stack (infra/amd/docker-compose.rocm.yml)
services:
  vllm-rocm: image: rocm/vllm:latest | devices: [/dev/kfd, /dev/dri] | shm_size: 16g
  command: vllm serve Qwen/Qwen2.5-7B-Instruct --host 0.0.0.0 --port 8000 --max-model-len 8192 --gpu-memory-utilization 0.9
```

Safety Invariants & Non-Goals

CARELYNX is an information navigation platform, not a doctor. We enforce strict clinical guardrails to eliminate autonomous medical decisions.

🚫 Strict Non-Goals (What CARELYNX Will NEVER Do)

- × **Never diagnose:** No inferring illnesses or interpreting lab results.
- × **Never alter dosages:** No prescribing or altering drug frequencies.
- × **Never pick a winner:** When records conflict, both are flagged for human review.
- × **Never hallucinate:** Items lacking exact verbatim quotes are discarded.
- × **Never replace clinicians:** Navigates paperwork, does not replace doctors.

🛡️ Multi-Tiered Safety Gates

- ✓ **Page Quality Gate (≥0.85):** Unreadable pages trigger human review.
- ✓ **Fact Confidence Gate (≥0.80):** Low confidence facts flagged `NEEDS_REVIEW` .
- ✓ **Verbatim Substring Linker:** Verifies every quote character-for-character.
- ✓ **Red-Flag Warning Banners:** Urgent emergency symptoms highlighted with ER alerts.
- ✓ **Zero-Downtime Rule Fallback:** Seamless fallback to deterministic rules if GPU is unreachable.

Multilingual Care Navigation

Empowering non-native speakers across 6 languages with our zero-risk Dosage Preservation Covenant.

Supported Languages

Full UI & Care Plan translation across:

ENGLISH | हिन्दी (HINDI) | ગુજરાતી (GUJARATI) | मराठी (MARATHI) | தமிழ் (TAMIL) | తెలుగు (TELUGU)

Dosage Preservation Rule

A life-saving safety covenant: **Brand drug names, chemical names, units (mg, mL, IU), and numerical frequencies remain untranslated** to prevent catastrophic transliteration errors that cause fatal overdoses.

Plain-Language Guidance

Converts clinical abbreviations and dense directives (e.g. *"ambulate with assistance", "NPO post midnight"*) into warm, culturally appropriate, actionable instructions in the patient's native tongue.

Clinical Safety Example: "Take Metformin 500mg with breakfast" → हिन्दी: "नाश्ते के साथ Metformin 500mg लें"

ZERO MEDICATION RISK

Dual-Persona Clinical Workflow

Empowering patients with clear daily guidance while giving clinical staff full audit oversight.

👤 Patient & Caregiver Experience

- ✓ **1-Click Upload:** Upload PDF discharge sheets or phone camera scans.
- ✓ **Interactive Daily Timeline:** Morning, afternoon, and evening medication schedule.
- ✓ **Click-to-Verify Citations:** Click any pill or appointment to see original document text.
- ✓ **Red-Flag Alerts:** Instant hospital return criteria highlighted with 24/7 hotline numbers.

👨‍⚕️ Clinical Review Workspace (`/review`)

- ✓ **RBAC Review Portal:** Restricted to authorized discharge nurses and reviewers.
- ✓ **Conflict Resolver:** Side-by-side view comparing conflicting documents.
- ✓ **Actionable Decisions:** APPROVE, REJECT, or MODIFY fact with resolution notes.
- ✓ **Immutable Audit Trail:** Every change recorded with actor type and timestamp.

Tested against 127 automated backend integration tests and synthetic clinical discharge scenarios.

PRODUCTION CERTIFIED

Hospital ROI & Healthcare Impact

Delivering measurable value for hospital operations, healthcare economics, and patient safety.

📉 Reduces Readmission Penalties

Hospitals lose millions under CMS Hospital Readmissions Reduction Program (HRRP). Clarifying discharge medication routines directly cuts avoidable 30-day readmissions.

📞 Cuts Nurse Triage Calls by 20-30%

Discharge nurses spend hours answering confused phone calls from families. CARELYNX provides immediate, authoritative answers directly cited to their paperwork.

🏥 Native FHIR R4 Interoperability

Exports verified medication facts into standard FHIR R4 `MedicationStatement` and `Patient` bundles, seamlessly synchronizing with Epic, Cerner, and hospital EHRs.

1-COMMAND SETUP

`./setup.sh` • `setup.bat` • `docker compose --profile amd up`

PRODUCTION HARDENED

NIST PBKDF2 + JWT Auth • IP Rate-Limiting • XSS/SQLi Sanitization

GROUNDED CLINICAL AI • AMD ROCM ACCELERATED

Empowering Every Patient to Recover with Clarity, Confidence & Dignity.

Thank you to AMD, Lablab.ai, and the healthcare community for championing open, private, and verifiable clinical AI.

TEAM
The Crawlers
AMD Developer Hackathon Act III

OPEN SOURCE GITHUB
github.com/v3da-bit/CARELYNX
Apache 2.0 / MIT Licensed

LIVE WEB & API
carelynx-two.vercel.app
Railway Backend Online